

Theories at Different Levels and Emergent Phenomena

- A theory may try to predict 'emergent phenomena' directly, without reducing it to the detailed behaviour of its constituent parts.
- A numerical model or a simulation is a calculation from which phenomena naturally emerge from the basic building blocks, maybe as a by-product. Very reductionist.

Emergent Phenomena - examples

- Temperature.
- The Gulf Stream, the Hadley Cell.
- Galaxies, clusters of galaxies, solar systems.
- Species in biological evolution. Consciousness.

Explicit Large Numerical Calculations

- Numerical calculations of colliding molecules, Maxwellian distributions.
- GCMs simulating the Gulf Stream and the Hadley Cell.
- Smoothed particle hydrodynamics for galaxies, cosmological N-body codes etc.
- [Species in evolutionary biology? Consciousness?] *Not yet!*